

13 Three statements about energy changes in chemical reactions are listed.

- 1 In an endothermic reaction, the temperature of the surroundings increases.
- 2 In an exothermic reaction, thermal energy is taken in from the surroundings.
- 3 In the reaction pathway diagram for an exothermic reaction, the energy level of the products is lower than the energy level of the reactants.

Which statements are correct?

- A** 1 and 2 **B** 1 only **C** 2 and 3 **D** 3 only

14 When a small piece of a Group I metal is placed into a large beaker of cold water, a reaction occurs.

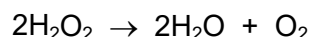
Four statements about this reaction are listed.

- 1 The metal melts.
- 2 Hydrogen is produced.
- 3 Steam is produced.
- 4 The pH of the solution increases.

Which statements about this reaction describe a physical change?

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

15 The equation for the decomposition of hydrogen peroxide is shown.



The reaction is exothermic.

When a small amount of a catalyst is added, the oxygen is produced more quickly.

Which statement about the catalyst is correct?

- A** The catalyst makes the reaction more exothermic.
- B** The mass of catalyst is the same before and after the reaction.
- C** The catalyst increases the final volume of oxygen produced.
- D** All of the catalyst is used up in the reaction.